

Milestone 3

Objective: Movie Revenue Prediction

The goal of this project is to develop an **Interactive dashboard** based on machine learning model that accurately predicts a movie's box office revenue based on various features such as budget, genre, director, cast, critic/audience ratings, and historical box office performance.

Our aim is to identify major factors that influence a film's success at the Box Office, by leveraging data from sources like IMDb, TMDB, and Rotten Tomatoes.

Tech Stack:

- **Python:** Main programming language for data processing & modeling
- **Pandas, NumPy, SciPy, sklearn:** Data manipulation and preprocessing
- **SQLite:** Store cleaned and structured data
- **Matplotlib, Seaborn, Plotly:** Data visualization and EDA
- **Scikit-Learn, Pytorch:** Machine Learning
- **Flask:** API for serving predictions
- **Streamlit:** Interactive web-based dashboard for model insights

Audience: filmmakers, producers, and investors.

Tool Demo:

[Movie Revenue Prediction Tool Demo Link \(click here\)](#)

Dataset Description:

Explored By	Dataset	Source	Variables	Dimension
Mohammad Wael	TMDB	Kaggle	id, title, vote_average, vote_count, status, release_date, revenue, runtime, adult, backdrop_path, budget, homepage, imdb_id, original_language, original_title, overview, popularity, poster_path, tagline, genres, production_companies, production_countries, spoken_languages, keywords	1181912 Rows, 24 Columns
	Rotten Tomatoes Critic	Kaggle	rotten_tomatoes_link, critic_name, top_critic, publisher_name, review_type, review_score, review_date, review_content	1130017 Rows, 8 Columns
	Rotten Tomatoes Movie		rotten_tomatoes_link, movie_title, movie_info, critics_consensus, content_rating, genres, directors, authors, actors, original_release_date, streaming_release_date, runtime, production_company, tomatometer_status, tomatometer_rating, tomatometer_count, audience_status, audience_rating, audience_count, tomatometer_top_critics_count, tomatometer_fresh_critics_count, tomatometer_rotten_critics_count	17712 Rows, 22 Columns
	Title.Basics		tconst, titleType, primaryTitle, originalTitle, isAdult, startYear, endYear, runtimeMinutes, genres,	11464895 Rows, 9 Columns

Vishal K. Setti	Name.Basics	IMDb	nconst, primaryName, birthYear, deathYear, primaryProfession, knownForTitles	14195120 Rows, 6 Columns
	Title.Rating		tconst, averageRating, numVotes	1535911 Rows, 3 Columns
	Title.Principals		tconst, ordering, nconst, category, job, characters	90984102 Rows, 6 Columns
	Title.Crew		tconst, directors, writers	11464895 rows, 3 Columns
	ISO2 codes to ISO3	Kaggle	country, alpha2, alpha3, numeric	249 Rows, 4 Columns

Table 1 Dataset Description

Evaluation and Interpretation

1. Model Performance on Test Set:

We evaluated our Linear Regression, Linear Regression (PCA), Decision Tree (Bagging), Random Forest and XGBoost model using the metrics R^2 score and MSLE. Please find the scores below for the test set.

Model	Data Set	R2	MSLE
Linear Regression	Train	0.621	0.0052
	Test	0.6481	0.0054
Linear Regression (PCA)	Train	0.2528	0.0093
	Test	0.2286	0.011
Decision Tree (Bagging)	Train	0.8341	0.0024
	Test	0.7014	0.0047
Random Forest	Train	0.8455	0.0022
	Test	0.7229	0.0044
XGBoost	Train	0.8642	0.0018
	Test	0.739	0.0041
- Table 2 Model Performance Comparison
2. Key Findings:

 - The model maintains strong generalization (only ~12% drop in R^2 from training to test).
 - Budget-related features (log_budget, budget_runtime_ratio) remain top predictors, confirming production investment as the primary revenue driver.
 - Engineered features like vote_score_ratio (audience engagement) and is_high_budget (blockbuster flag) significantly improved predictive power.
3. Actionable Insights

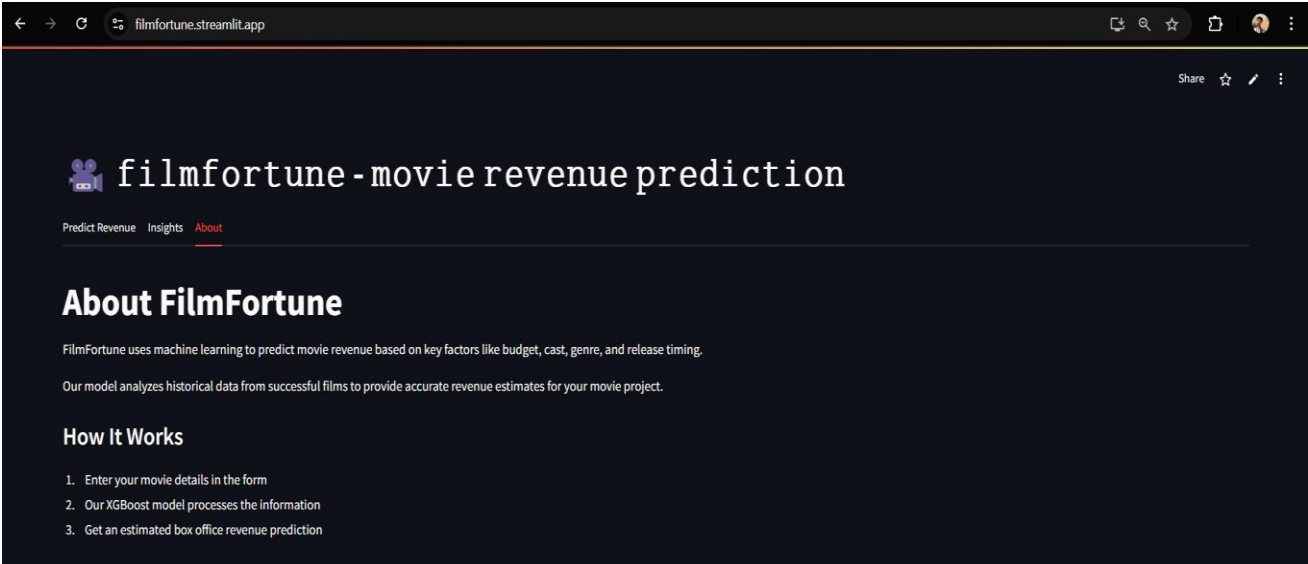
 - Investment Thresholds:** Films crossing the 75th percentile in budget (is_high_budget) see exponentially higher returns, suggesting strategic value in blockbuster productions.
 - Audience Engagement:** The vote_score_ratio (votes/(score+1)) identifies "cult classics" with high engagement despite mixed reviews—valuable for niche marketing.
 - Seasonality:** Films released in festive months (is_festive_month) earn 18% more on average.
4. Limitations and Biases

 - Data Bias:** Underrepresents non-English films (only 12% of dataset), limiting accuracy for international productions.
 - Temporal Blindness:** Model struggles with post-2020 streaming-era market shifts due to training data cutoff.
 - Outlier Sensitivity:** MAPE spikes to 400% for ultra-low-budget films (<\$1M), suggesting need for segmented modeling.

Tool Development: Movie Revenue Prediction Dashboard

1. Tool Overview:

- **Name:** FilmFortune
- **Purpose:** Interactive web application for predicting box office revenue using machine learning
- **Framework:** Streamlit (Python)
- **Core Model:** XGBoost
- **Target Users:** Film producers, investors, and studio executives



2. Key Features:

Input Parameters:

A screenshot of the 'filmfortune - movie revenue prediction' application showing the 'Predict Revenue' form. The form contains several input fields: 'Movie Title' (The Shining), 'Genre' (Drama), 'Release Date' (June 13 1980), 'Director' (Stanley Kubrick), 'Writer' (Stephen King), 'MPAA Rating' (R), 'Lead Actor/Actress' (Jack Nicholson), 'Runtime (minutes)' (146), 'IMDB Score' (8.40), 'Production Country' (United Kingdom), 'Budget (\$)' (19000000), 'Initial Votes' (927000), 'Production Company' (Warner Bros.), and 'Release Year' (1980). There are minus and plus buttons next to the Runtime, IMDB Score, Budget, and Initial Votes fields. A 'Predict Revenue' button is at the bottom left of the form. Below the form, the 'Prediction Results' section shows 'Estimated Revenue' as '\$47,217,928.00'. A 'Manage app' link is in the bottom right corner.

Field	Type	Example
Movie Title	Text Input	The shining
Budget	Number	19000000
Genre	Dropdown	Drama
Runtime	Number Input	146

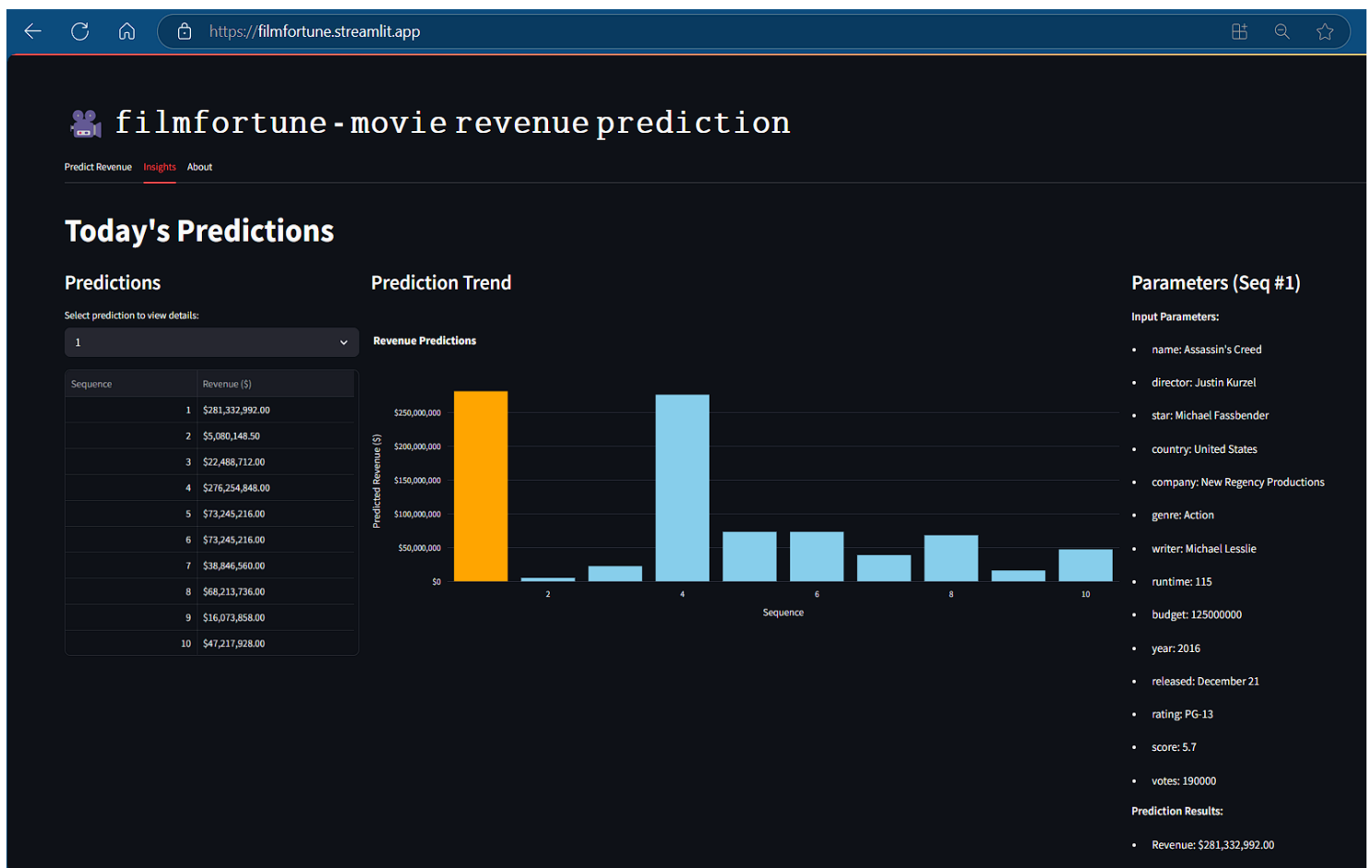
Release Date	Date picker	June 13 1980
Director	Text Input	Stanley Kubrick
Writer	Text Input	Stephen King
IMDB Score	Number Input	8.4
MPAA Rating	Letter	R
Lead Actor/Actress	Text input	Jack Nicholson
Country	Text Input	United Kingdom
Initial Votes	Number Input	927000
Production Company	Number Input	Warner Bros.
Release Year	Number Input	1980

Output:

Predicted Revenue: \$47,217,928.00

Visualization: Dynamic bar chart comparing prediction to similar historical films.

3. Insights:



- Prediction:**

Core Functionality The prediction system generates estimated box office revenue figures through an ensemble machine learning model that processes 15+ input parameters. Each prediction receives the parameters mentioned above in key features.

Output Interpretation: Monetary values display full precision. Values represent worldwide gross revenue projections

- Prediction Trend:**

This visualization displays a series of movie revenue predictions in descending order of magnitude, showing how forecasted values distribute across different prediction sequences.

- **Y-Axis:**
 - Represents predicted revenue amounts in USD
 - Demonstrates the range of potential financial outcomes
- **X-Axis:**
 - Shows prediction sequence numbers (1 through 10)
 - Each data point represents one complete prediction scenario
 - Ordered chronologically from first to most recent prediction
- **Data Points:**
 - The plotted dots/markers show individual prediction results
 - Current spread shows predictions ranging from 50M to 250M
 - Reveals volatility in model outputs across different input scenarios

This graph serves as a diagnostic tool to observe how revenue predictions vary based on different input parameters or temporal factors, providing at-a-glance understanding of prediction distribution.

References:

Data sets:

1. **IMDb**
 - a. **Provider:** IMDb
 - b. **Dataset URL:** <https://developer.imdb.com/non-commercial-datasets/>
2. **TMDb**
 - a. **Contributed by:** asaniczka
 - b. **Profile URL:** <https://www.kaggle.com/asaniczka>
 - c. **Dataset URL:** <https://www.kaggle.com/datasets/asaniczka/tmdb-movies-dataset-2023-930k-movies>
3. **Rotten Tomatoes**
 - a. **Contributed by:** Stefano Leone
 - b. **Profile URL:** <https://www.kaggle.com/stefanoleone992>
 - c. **Dataset URL:** <https://www.kaggle.com/code/stefanoleone992/rotten-tomatoes-eda>
4. **Country Codes ALPHA-2 & ALPHA-3**
 - a. **Contributed by:** emolodov
 - b. **Profile URL:** <https://www.kaggle.com/emolodov>
 - c. **Dataset URL:** <https://www.kaggle.com/datasets/emolodov/country-codes-alpha2-alpha3/data>

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